

Kartikeya Gaur

kartikeyagaur9@gmail.com | [LinkedIn](#) | [Website](#) | [GitHub](#)

EDUCATION

Indian Institute of Science (IISc) <i>M.Tech. in Smart Mobility and Logistics Systems</i>	Bengaluru, India 2027
National Institute of Technology (NIT) <i>B.Tech. in Civil Engineering</i>	Hamirpur, India 2024

EXPERIENCE

Research Intern <i>University of New South Wales (UNSW)</i>	Jun – Jul 2026 <i>Sydney, Australia</i>
Developed a comparative public transit route-choice modeling framework for Sydney and Bengaluru, implementing Expanded Path Size and Nested Recursive Logit using real GTFS and synthetic ridership data. [Report] [GitHub]	
Junior Research Fellow <i>Indian Institute of Remote Sensing (IIRS), ISRO</i>	Sept 2024 – July 2025 <i>Dehradun, India</i>
- Automated bathymetry and discharge estimation in data-scarce river basins using ICESat-2 LiDAR altimetry (DBSCAN, Monte Carlo), achieving ~6% deviation from field discharge measurements at Varanasi. - Led a technical report on wide-swath altimetry and mentored two interns on SWOT and FF-SAR. [Report]	

PROJECTS

Scalable Algorithm for Public Transit Skims	<i>M.Tech Thesis</i>
Contributing to a scalable C++ routing framework designed to generate transit network impedance matrices, aiming to simulate millions of agent journeys for Bengaluru.	
Warehouse Optimization at Land Port Petrapole	<i>Industry Capstone with Ernst & Young (EY) LLP</i>
- Working along a 5-person team over a semester, studied Petrapole warehouses at 191.7% of designed capacity via a 5S audit and System Dynamics model, highlighting its core operational inefficiencies. - Proposed a phased Action Plan including Selective Pallet Racking + ABC-XYZ zonation plus a 5-layer IoT architecture netting INR 1.02 Cr/yr against INR 41 L cost. [Report]	
Nighttime Street Light Audit Using Computer Vision	<i>YOLOv8, PyTorch</i> <i>Edge AI</i>
- Working along a 4-person team, collected 16.6 km of video and hand-labeled ~2.1k frames for streetlights. - Shrank the model with 2.5 million parameters for on-device hardware up to 50% sparsity exporting to ONNX/TFLite-INT8 for real-time (3.2 ms) smartphone inference in a deployable web app. [GitHub] [Demo]	
Intra-Urban Heat Island (UHI) Effect	<i>Google Earth Engine, QGIS</i> <i>B.Tech Project</i>
Processed 20 years (2003–2023) of Landsat-8 thermal imagery to quantify UHI effect; Found a 2.80°C rise in Chandigarh (planned) (0.14°C/yr) versus a sharper 6.77°C rise in Jalandhar (unplanned) (0.34°C/yr). [Report]	

TECHNICAL SKILLS

Languages: Python, C++, R, Julia
Geospatial: ArcGIS, QGIS, Remote Sensing, Google Earth Engine, OpenStreetMap, GTFS
Machine Learning: TensorFlow, Keras, PyTorch, Scikit-learn; model compression; LLM/RAG inference;
Scalable Computing: Apache Spark*, Parallel & Distributed Computing*
Tools & BI: Power BI, Jupyter Notebook, Git, GitHub
Coursework: Multi-Modal Mobility Systems; Network Science and Optimization; ML for Cyber-Physical Systems

PUBLICATIONS

- Gaur, K., et al. (2025). “SWOT Mission with Wide-Swath Altimetry: Observations and Insights into India’s Inland Waterbodies.” *Journal of the Indian Society of Remote Sensing*.

ACHIEVEMENTS & EXTRA-CURRICULAR

- Secured All India Rank (AIR) 10** in GATE 2023 (Geomatics Engineering) with an 819 score. [Score Card]
- Student Auditor**, Literacy Mission, NITH: managed finances for 80+ children; raised ~INR 5.5 Lakh in FY23–24.

* Currently learning through DS221: Introduction to Scalable Systems, IISc.